

INDIAN STATISTICAL INSTITUTE

Project Linked Person — Selection 2026–2027

Subject: Systems Support (Software, Hardware and Web)

Part B — Practical Examination

Total: 16 + 18 + 16 = 50 marks Duration: 90 minutes

EXAMINATION INSTRUCTIONS

1. You have root on a virtual machine. It is a disposable clone: you cannot damage anything that matters, and it is connected to no real network. Log in with `ssh candidate@192.168.56.10`.
2. Each question has four parts — *the problem, how to reproduce it, what to fix, and how to check your work*. Run the checks yourself; you should not have to guess whether you are finished.
3. **The *Constraints* of every question are marked. Breaking a constraint loses marks even when the symptom goes away. Examples: `chmod 777`, pinning names into `/etc/hosts`, reformatting a filesystem.**
4. Keep `~/FIXLOG.md` up to date as you go: symptom, cause, what you changed, and how you verified it. It is marked, and there is a short oral discussion of your work afterwards.
5. You are not expected to know all of this in advance. Partial credit is real: a fault you diagnose correctly and explain, but cannot finish, is worth more than one you paper over.

NOTE: `man`, `--help`, `journalctl` and the logs under `/var/log` are all available on the machine. Your commands are recorded. Questions may be attempted in any order.

Q1. Disk full on the lab file server.

[16 marks]

Reported by a research scholar: “I cannot copy anything into `/srv/data` — it says the disk is full, and nothing I can see in there accounts for that much space. Submitting a job into `/srv/spool` *also* says the disk is full, but `df -h` shows that one has plenty of space left. That part makes no sense to me.”

Two storage faults, on two different filesystems. `/srv/data` is genuinely full, and the files you can see do not add up to the space reported as used. `/srv/spool` refuses to create files *although* `df` reports free space — something other than free bytes has run out.

How to reproduce it

<code>df -h /srv/data</code>	# 100% used
<code>df -h /srv/spool</code>	# shows space free
<code>touch /srv/spool/incoming/testjob</code>	# "No space left on device"
<code>sudo du -sh /srv/data</code>	# smaller than df says is in use

What to fix

- 1.1 Bring `/srv/data` back to **under 10% used** (the `Use%` column of `df -h`). Freeing the largest obvious files will not get you there on its own. Treat any difference

between what `df` says is used and what `du` can account for as something you still have to explain.

- 1.2 Get the reporting daemon writing to `/srv/data/logs/reportd.log` again. It is running; its writes were failing. Check the last line's timestamp rather than assuming.
- 1.3 Make `/srv/spool` able to accept new files. It is a separate filesystem with a separate fault and it is *not* short of space: freeing bytes there will not help. A filesystem can exhaust more than one kind of resource, and `df` has another mode that reports the other one.
- 1.4 In `~/FIXLOG.md`, explain in two or three sentences why `/srv/spool` reported "No space left on device" while `df` showed free space.

How to check your work

```
df -h /srv/data                # Use% under 10%
touch /srv/spool/incoming/t && echo OK && rm /srv/spool/incoming/t
ls /srv/data/current           # five survey-block CSV files, still there
```

Constraints — these are marked

- Nothing under `/srv/data/current` may be deleted or changed. It is live survey data with no second copy on this machine.
- Do not reformat, recreate, resize or unmount either filesystem.
- `/srv/data/archive/README.txt` says what is safe to remove there. Read it before deleting anything.
- Rebooting clears one of these faults. On a real file server at 11am you would not have that option; treat it as unavailable.

Q2. Shared project folder unusable. [18 marks]

Reported for the Statistics Lab group: "Bikram cannot save anything into `shared/`, and cannot open `notes.md` at all. When I create a file there, the others cannot edit it — it comes out belonging to my own personal group instead of the project group. `sudo systemctl restart reportd` used to work for all of us; now it is refused. And Chandan from the internal audit cell needs to *read* that folder for the annual review. He cannot see anything."

`/srv/projects/statlab/` is a shared area for the `statlab` group. Several things about it have been changed, and the auditor has no access at all.

How to reproduce it

```
sudo -u bikram touch /srv/projects/statlab/shared/test      # Permission denied
sudo -u bikram cat /srv/projects/statlab/shared/notes.md    # Permission denied
sudo -u anita sudo -n systemctl restart reportd            # refused
sudo -u chandan cat /srv/projects/statlab/shared/notes.md  # Permission denied

stat -c '%a %U:%G %n' /srv/projects/statlab
id bikram ; getfacl /srv/projects/statlab
sudo cat /etc/sudoers.d/statlab
```

What to fix

- 2.1 Everyone in `statlab` can read and write everything under `/srv/projects/statlab/`.
- 2.2 A file created there by one member ends up usable by the whole group.
- 2.3 `notes.md` is a shared file again, not one person's private file.
- 2.4 Members of `statlab` — and only them — can run `sudo -n systemctl restart reportd` without a password.
- 2.5 Chandan can **read** that tree.

How to check your work

Test as the affected users, not as root.

```
sudo -u bikram touch /srv/projects/statlab/shared/t1 && echo "write OK"
stat -c '%U:%G' /srv/projects/statlab/shared/t1    # group must be: statlab
sudo rm -f /srv/projects/statlab/shared/t1
sudo -u bikram cat /srv/projects/statlab/shared/notes.md >/dev/null && echo "read OK"
sudo -u anita sudo -n systemctl restart reportd && echo "sudo OK"
sudo -u chandan cat /srv/projects/statlab/shared/notes.md >/dev/null && echo "auditor read OK"
sudo -u chandan touch /srv/projects/statlab/shared/t2 && echo "PROBLEM: auditor can write"
```

The second line is the one people miss. If the group comes out as anything other than `statlab`, the second complaint is not fixed yet.

Constraints — these are marked

- **Chandan must not be put into the `statlab` group.** Audit policy forbids putting auditors into the groups they audit. He must be able to read and must *not* be able to write.
- No `chmod 777`, anywhere. This is a multi-user machine.
- Do not change anyone's login shell, password or home directory.

Hint. One of these faults is about which **bits are set on a directory**, not about who owns it — a directory carries more than the nine permission bits you see first. Group membership changes do not affect a shell that is already open. Edit anything under `/etc/sudoers.d/` with `visudo -f`.

Q3. Campus names stopped resolving. [16 marks]

Reported by the systems group: “This host cannot resolve any campus name since the patching window. `ping www.isi.local` sits there for several seconds and then fails. The odd one is `portal.isi.local` — that *does* answer, but with an address that has not been ours since 2023. Please do not paper over this with hosts-file entries; we have forty more machines behind this one.”

The host runs its own resolver (**dnsmasq**) serving the `isi.local` zone and forwarding everything else upstream. The addresses it is supposed to hand out:

name	address
<code>www.isi.local</code>	127.0.0.1
<code>statlab.isi.local</code>	127.0.0.1
<code>portal.isi.local</code>	127.0.0.1
<code>git.isi.local</code>	10.10.10.9
<code>nas.isi.local</code>	10.10.10.20

How to reproduce it

```
getent hosts www.isi.local      # nothing, after a long pause
getent hosts git.isi.local     # nothing
getent hosts portal.isi.local  # answers 192.0.2.77 -- wrong address

dig +short @127.0.0.1 git.isi.local
systemctl status dnsmasq
```

What to fix

- 3.1** All five names resolve to the correct addresses, for every user on the host, in under a second.
- 3.2** The answers come from the resolver, not from a local file.
- 3.3** It still works after a reboot.

How to check your work

```
for n in www statlab portal git nas; do
  printf '%-22s %s\n' "$n.isi.local" "$(getent hosts $n.isi.local | awk '{print $1}')"
done
dig +short @127.0.0.1 git.isi.local      # must print 10.10.10.9
grep -i isi.local /etc/hosts             # must print nothing at all
systemctl is-active dnsmasq ; systemctl is-enabled dnsmasq
```

Constraints — these are marked

- **Do not put isi.local names into /etc/hosts.** The systems group will not maintain hosts files across forty machines. It makes the symptom disappear and loses marks.
- Do not make /etc/resolv.conf immutable (chattr +i) to hold a fix in place.

Hint. There is more than one fault and they are in different layers. A useful order to think in: *is a DNS query being sent at all? where is it being sent? is anything answering? is the answer correct?* A service that refuses to start will usually say why, in its own log. The long pause before failure is itself a clue about which layer is wrong.